

AY25 PROGRAM ASSESSMENT REPORT

v2026.1

Tracking Data

CIM code (e.g., ME-CM)	MS-ASAR
Program Name	Master of Science in Autonomous Systems and Robotics
Assessment Year	2025
Program Assessment Coordinator	Heng Wang
Year of Current Assessment Plan	2024
No Assessment This Year per Plan	☐
LO(s) Assessed	LO1, LO2
Presented at program authority meeting date: <i>Please attach minutes</i>	TBD

Methodology

LO#1

- ⌘ Learning goal assessed: Graduates will be able to apply computational and numerical techniques to solve relevant engineering problems.
- ⌘ Sample size: All students completing MMAE 501
- ⌘ Semesters in which artifacts were collected: Fall, Spring
- ⌘ Artifact: For each course, performance of the students in 4 questions chosen from a pool of questions specifically designed to assess this learning goal. The questions are delivered to the students in the last third of the semester. Each student will be scored out of 4.
- ⌘ Proficiency ratings
 - ⌘ scores of >75% points indicate **P** proficient.
 - ⌘ scores of >55% points indicate **M** meets expectations.
 - ⌘ scores below 55% points indicate **D** does not meet expectations.

LO#2

- ⌘ Learning goal assessed: Graduates will demonstrate solid knowledge in autonomous systems and robotics.
- ⌘ Sample size: All students completing a subset of MMAE 541, MMAE 543
- ⌘ Semesters in which artifacts were collected: Fall, Spring
- ⌘ Artifact: For each course, performance of the students in 4 questions chosen from a pool of questions especially designed to assess this learning goal. The questions are delivered to the students in the last third of the semester. Each student will be scored out of 4.

- ∄ Proficiency ratings
 - ∄ scores of >75% points indicate **P** proficient.
 - ∄ scores of >55% points indicate **M** meets expectations.
 - ∄ scores below 55% points indicate **D** does not meet expectations.

LO#3

- ∄ Learning goal assessed: For graduates doing thesis, they will have the ability to conduct research, publish scientific findings, and disseminate that knowledge.
- ∄ Sample size: All students completing their MS thesis.
- ∄ Semesters in which artifacts were collected: Fall, Spring
- ∄ Artifact: A specially designed questionnaire filled out by the thesis committee after the thesis defense exams. The committee members come to a consensus and rate the students on a scale of 1-5 on each of the following:
 - ∄ Research problem is clearly described
 - ∄ Literature and context of the work are demonstrated
 - ∄ Solution approach, methods and/or tools are described
 - ∄ Results are analyzed, interpreted and communicated well
 - ∄ Written and oral communication are effective
 - ∄ Critical thinking consistent with an independent researcher
- ∄ Proficiency ratings
 - ∄ scores of 4 points and above for all items indicate **P** proficient.
 - ∄ scores of 3 points and above for all items indicate **M** meets expectations.
 - ∄ scores below 3 points on any item indicate **D** does not meet expectations.

Results

Artifact samples

Artifact for LO1 from MMAE 501. Four questions from a bank of 10 were selected and assigned as a quiz. Sample scores listed below. "C" for correct, "I" for incorrect.

1
Multiple Choice
1 point

Given a system of linear differential equations $\frac{dx}{dt} = Ax$, where A is a square matrix, which statement is true about the general solution?

- ☒ If A has repeated eigenvalues, the solution may involve terms of the form $t^k \cdot e^{\lambda t}$
- ☐ It always contains n linearly independent solutions, where n is the size of A
- ☐ It can be expressed as $x(t) = e^{At}x(0)$ only if A is invertible
- ☐ The stability of the system depends solely on the determinant of A

MS-MAE LO1 MS-ASAR LO1

2
Multiple Choice
1 point

In the calculus of variations, what is the primary difference between constrained and unconstrained optimization problems?

- ☒ Constrained problems require the use of Lagrange multipliers
- ☐ Constrained problems always lead to simpler Euler-Lagrange equations
- ☐ Unconstrained problems can only be solved using direct methods
- ☐ The extremals for constrained problems are always smooth curves

MS-MAE LO1 MS-ASAR LO1

3
Multiple Choice
1 point

Consider the Sturm-Liouville problem: $-(p(x)y')' + q(x)y = \lambda r(x)y$ with appropriate boundary conditions. Which statement is correct?

- ☒ The eigenfunctions form an orthogonal set with respect to the weight function $r(x)$
- ☐ The eigenvalues are always real, regardless of the functions $p(x)$, $q(x)$, and $r(x)$
- ☐ The number of nodes in the n th eigenfunction is always exactly n
- ☐ The lowest eigenvalue is always positive if $q(x) > 0$ for all x

MS-MAE LO1 MS-ASAR LO1

4
Multiple Choice
1 point

For a complex function $f(z) = u(x,y) + iv(x,y)$, which condition ensures that $f(z)$ is analytic?

- ☒ $u_x = v_y$ and $u_y = -v_x$ (where u_x is the partial derivative of u w.r.t x etc.)
- ☐ $u_x = v_y$ and $u_y = v_x$
- ☐ $u^2 + v^2 = \text{constant}$
- ☐ $u_{xx} + u_{yy} = 0$

MS-MAE LO1 MS-ASAR LO1

	Sample 1		
P	C, C, C, C (100%)		
M	C, I, C, C (75%)		
D			

Artifact for LO2 from MMAE 543. Four questions from a bank of five were selected and assigned as a quiz. Sample scores listed below. "C" for correct, "I" for incorrect.

1
Multiple Choice
1 point

In the context of Kalman filtering, which of the following statements is true?

- ☒ The filter provides the optimal estimate in the minimum mean square error sense
- ☐ The Kalman gain is constant for time-invariant systems
- ☐ The filter performance is independent of the assumed process noise covariance
- ☐ The estimate error covariance always decreases monotonically with time

MS-ASAR LO2

2
Multiple Choice
1 point

Given a discrete-time system described by $x(k+1) = Ax(k) + Bu(k)$, which of the following statements about its controllability is correct?

- ☒ The system is controllable if and only if $\text{rank}[B \ AB \ A^2B \ \dots \ A^{n-1}B] = n$, where n is the system order.
- ☐ Controllability is independent of the sampling time.
- ☐ A system controllable in continuous-time is always controllable in discrete-time.
- ☐ Controllability can be determined solely from the eigenvalues of A .

MS-ASAR LO2

3
Multiple Choice
1 point

Consider a continuous-time dynamic system $\dot{x} = Ax + Bu$, $y = Cx$. Which of the following statements about its discrete-time equivalent, with sampling interval T , is correct?

- ☒ Zero-order hold discretization introduces an effective time delay of $T/2$.
- ☐ The discrete-time A matrix is always equal to e^{AT} , even if A is time variant.
- ☐ The observability of the system is preserved regardless of the sampling time.
- ☐ The discrete-time B matrix is always equal to BT .

MS-ASAR LO2

4
Multiple Choice
1 point

For a Linear Quadratic Regulator (LQR) design, what is the effect of increasing the state cost matrix $Q1$ while keeping the control cost matrix $Q2$ constant?

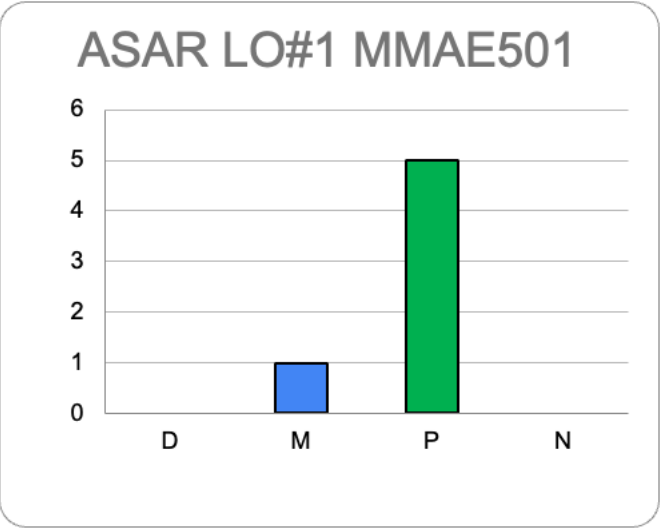
- ☒ It leads to more aggressive control actions.
- ☐ It results in slower system response.
- ☐ It minimizes actuator effort.
- ☐ It can lead to instability.

MS-ASAR LO2

	Sample 1	Sample 2	
P	C, C, C, C (100%)	C, C, C, C (100%)	
M	C, C, I, C (75%)	I, C, C, C (75%)	
D			

Data table

Charts below show the distribution of results from different courses. Y axis is the number of students.





Description of findings

This is the first year with the current assessment plan hence no comparison with previous years is available. The small student body has led to small numbers of sample points for assessment. The department believe a meaningful analysis of the results is only warranted when sufficient data points are collected (say >20). Based on what is available this year, no particular concern was identified. ~80% of students scored a perfect score in the assessment of their basic numerical and computation skills. Although the total number of students is only 7, we notice this percentage is comparable to the overall percentage in that course, reflecting a leveled input and outcome in terms of students. The assessment on learning objective #2 returned a slightly different picture through MMAE 543, with ~50% students scoring perfectly and ~50% missing one out of four questions. The total number of students assessed is only 11, making it hard to tell if this indicated any deficiency. The department will keep monitoring this and validating its indication by including more courses to assess this learning objective.

Continuous Improvement, CI, plans

The purpose of assessment is to improve the quality of our educational programs. The most important part of the assessment cycle is how we use the data to improve our programs. Examine the data provided above describing student achievement, consider the root cause of any deficits, and if areas for possible improvement emerge, describe plans to achieve this improvement.

Prior CI plan impact

Also, please refer to the previous assessment report for the LOs assessed. Were any CI plans recommended? Were they implemented? Did the plans work (were improvements seen in this cycle)?

☒ no previous CI plans available

This cycle's CI plan

Describe any CI plans developed from this cycle's data, with

- Specific curricular or instructional changes in response to this assessment data,
- A timeline for implementation,
- As well a description of who will participate in and manage these plans

☐ After consideration and presentation to the program authority, no CI plans are recommended.

1. Expand current assessment plan to include more courses.
2. While current assessment format with standardized questions is satisfactory, improvements are still preferred including: i) expanding the question banks and assigning more questions. Some of these can probe students' case-specific problem-solving skills, like which method should be used to better solve a problem. ii) collecting additional artifacts from homework or exams that solve specific problems.
3. Specifically on LO#2, the department will monitor assessment results from MMAE 543, as well as use more courses to assess the same learning outcome.